

01

a) $\frac{dy}{dt} = (y-\alpha)(y-\beta)$

when

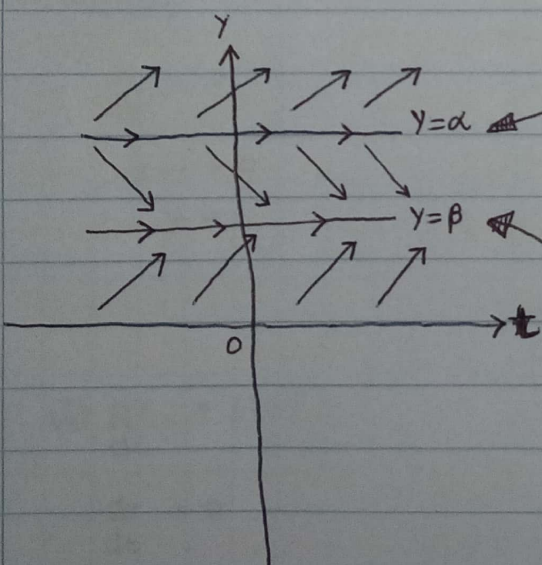
i) $\frac{dy}{dt} = 0$, we get critical points.

$$(y-\alpha)(y-\beta) = 0$$

$$y = \alpha, y = \beta$$

ii)

if $\alpha > \beta$



* Directional fields away from the critical point. Hence "unstable".

* Directional fields towards the critical point. Hence "stable".

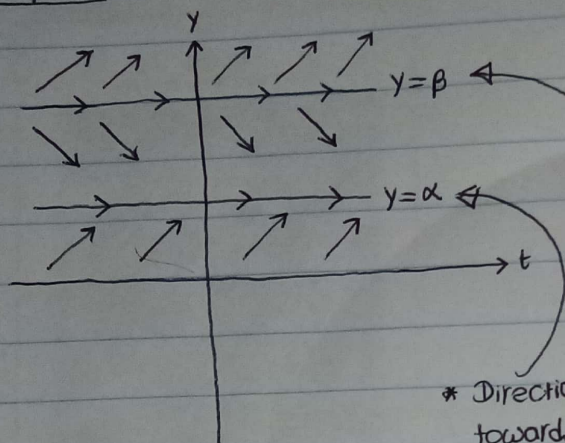
$$\frac{dy}{dt} = (y-\alpha)(y-\beta)$$

$$y > \alpha, \frac{dy}{dt} > 0$$

$$\beta < y < \alpha, \frac{dy}{dt} < 0$$

$$y < \beta, \frac{dy}{dt} > 0$$

if $\beta > \alpha$



$$\frac{dy}{dt} = (y-\alpha)(y-\beta)$$

* Directional fields away from the critical point. Hence "unstable"

* Directional fields towards the critical point. Hence "stable"

(b)

$$(i) \quad y^2 + \left(\frac{dy}{dx}\right)^2 = 1$$

$$y^2 + \left(\frac{dy}{dx}\right)^2 = 1$$

$$\left(\frac{dy}{dx}\right)^2 = 1 - y^2$$

$$\frac{dy}{dx} = \pm \sqrt{1 - y^2}$$

$$\frac{dy}{dx} = \sqrt{1 - y^2} \quad \text{or} \quad \frac{dy}{dx} = -\sqrt{1 - y^2}$$

$$\frac{1}{\sqrt{1 - y^2}} dy = 1 dx \quad \text{or} \quad -\frac{1}{\sqrt{1 - y^2}} dy = 1 dx$$

$$\int \frac{1}{\sqrt{1 - y^2}} dy = \int 1 dx \quad \text{or} \quad -\int \frac{1}{\sqrt{1 - y^2}} dy = \int 1 dx$$

$$\sin^{-1} y = x + C \quad \text{or} \quad -\sin^{-1} y = x + C \quad (C \text{ arbitrary constant})$$

$$y = \sin(x + C) \quad \text{or} \quad y = \sin(-x + C)$$

$$(ii) \frac{dy}{dx} = \frac{y}{x} - \frac{x}{y}$$

$$\frac{dy}{dx} = \frac{(y^2 - x^2)}{xy}$$

Let's check for whether, we have homogeneous D.E.

$$y \rightarrow yt$$

$$x \rightarrow xt$$

$$f(x, y) = \frac{y^2 - x^2}{xy}$$

$$f(xt, yt) = \frac{(yt)^2 - (xt)^2}{xt \cdot yt} = \frac{(y^2 - x^2)t^2}{xyt^2} = \frac{y^2 - x^2}{xy} = f(x, y)$$

$$\therefore f(x, y) = f(xt, yt)$$

$\therefore$ Homogeneous

So, we use $y = vx$ substitution,

$$\frac{d(vx)}{dx} = \frac{(vx)^2 - x^2}{x \cdot vx} = \frac{(v^2 - 1)x^2}{vx^2} = \frac{v^2 - 1}{v}$$

$$v \times 1 + x \frac{dv}{dx} = \frac{v^2 - 1}{v}$$

$$x \frac{dv}{dx} = \frac{v^2 - 1}{v} - v$$

$$x \frac{dv}{dx} = \frac{v^2 - 1 - v^2}{v} = \frac{-1}{v}$$

$$x \frac{dv}{dx} = \frac{-1}{v}$$

$$\frac{dv}{dx} = \frac{-1}{vx}$$

$$V dv = -\frac{1}{x} dx$$

$$\int V dv = \int -\frac{1}{x} dx$$

$$\frac{V^2}{2} = -1 \ln|x| + C \quad (C \text{ is an arbitrary constant})$$

$$V^2 = -2 \ln|x| + C$$

$$y = vx \rightarrow v = \frac{y}{x}$$

$$\frac{y^2}{x^2} = -2 \ln|x| + C$$

$$y^2 = -2x^2 \ln|x| + Cx^2$$

$$y^2 = x^2 [-2 \ln|x| + C]$$

$$y^2 = x^2 [\ln(x^{-2}) + \ln C]$$

$$y^2 = x^2 \ln(Cx^{-2})$$

$$y^2 = x^2 \ln \left| \frac{C}{x^2} \right| \quad (C \text{ is arbitrary constant})$$

$$y = \pm \sqrt{x^2 \ln \left| \frac{C}{x^2} \right|}$$

$$\begin{aligned} &\rightarrow y = x \sqrt{\ln \left| \frac{C}{x^2} \right|} \\ &\text{or} \\ &\rightarrow y = -x \sqrt{\ln \left| \frac{C}{x^2} \right|} \end{aligned}$$

$$(c) \quad \frac{dy}{dx} + \frac{e^x y + 2x}{2y + e^x} = 0, \quad y(0) = 0$$

$$\frac{dy}{dx} = -\frac{(e^x y + 2x)}{(e^x + 2y)}$$

$$\underbrace{(e^x y + 2x)}_{M(x,y)} dx + \underbrace{(e^x + 2y)}_{N(x,y)} dy = 0$$

Let's check for exactness,

$$\frac{\partial M}{\partial y} = e^x, \quad \frac{\partial N}{\partial x} = e^x$$

Since $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$, we have an exact D.E.

So, there exists a function $f(x,y)$ such that, $df = M dx + N dy$

$$df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy$$

$$\frac{\partial f}{\partial x} = M$$

$$\frac{\partial f}{\partial y} = N$$

$$\frac{\partial f}{\partial x} = e^x y + 2x \quad \text{--- (1)}$$

$$\frac{\partial f}{\partial y} = e^x + 2y \quad \text{--- (2)}$$

①,

$$f = \int (e^x y + 2x) dx$$

$$f = ye^x + 2\frac{x^2}{2} + h(y) \quad (h(y) \text{ is a function of } y.)$$

$$f = ye^x + x^2 + h(y)$$

$$f(x, y) = ye^x + x^2 + h(y) \text{ ——— ③}$$

③,

$$\frac{\partial f}{\partial y} = e^x + 0 + h'(y)$$

$$\frac{\partial f}{\partial y} = e^x + h'(y) \text{ ——— ④}$$

③ = ④,

$$e^x + 2y = e^x + h'(y)$$

$$h'(y) = 2y$$

$$h(y) = y^2$$

$$\textcircled{3}, \quad f(x, y) = ye^x + x^2 + y^2$$

But, $df = 0 \rightarrow f$ is a constant

$$\therefore ye^x + x^2 + y^2 = c \quad (c \text{ is an arbitrary constant})$$

I.V.P.

$$x=0, y=0, \quad 0e^0 + 0 + 0 = c$$

$$c = 0$$

$$ye^x + x^2 + y^2 = 0$$

$$ye^x + x^2 + y^2 = 0$$

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$$y^2 + e^x y + x^2 = 0$$

$$y = \frac{-e^x \pm \sqrt{e^{2x} - 4x^2}}{2}$$

$$y = \frac{-e^x \pm \sqrt{e^{2x} - 4x^2}}{2}$$

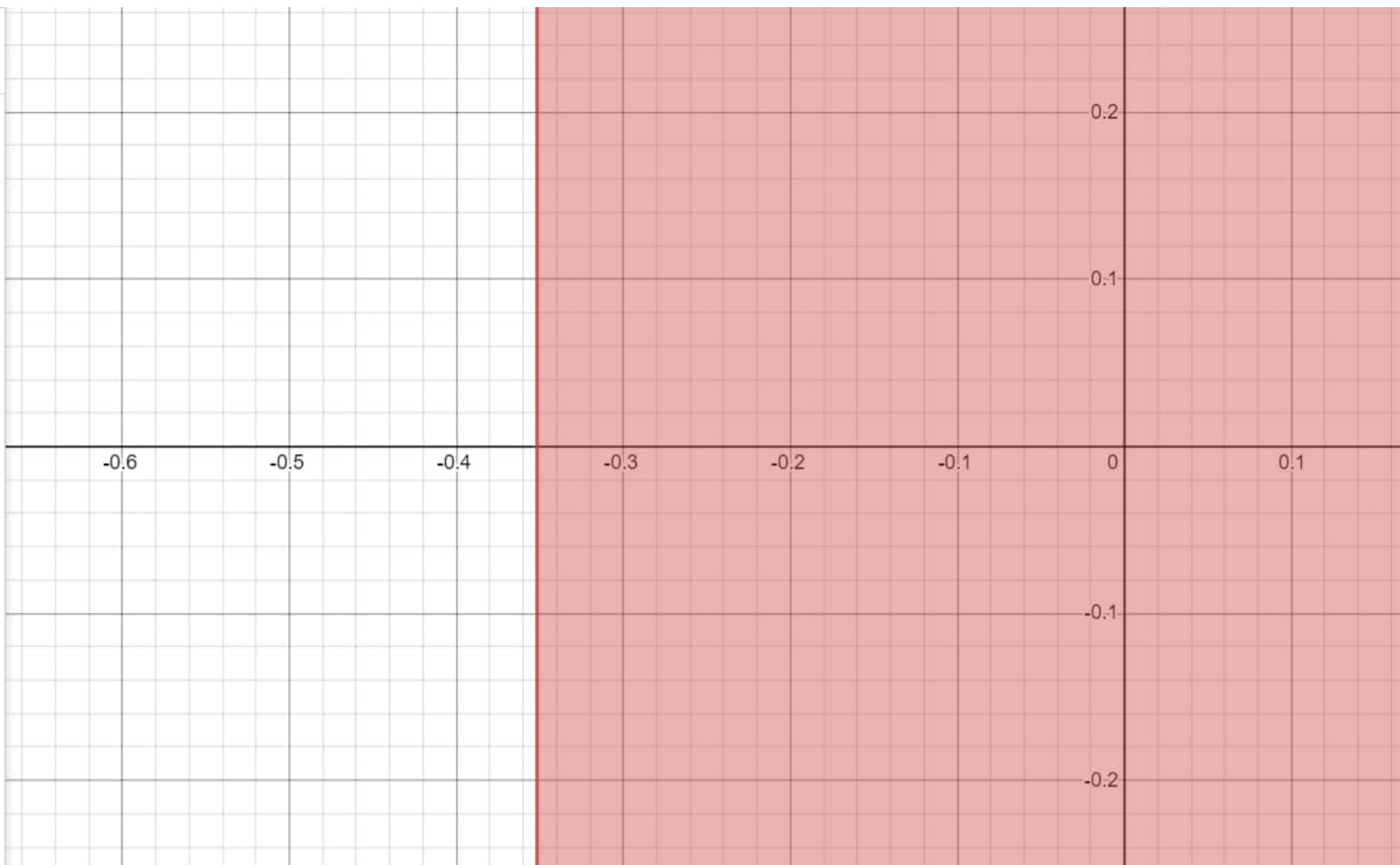
To ensure that y remains real,

$$e^{2x} - 4x^2 \geq 0$$

$$(e^x)^2 - (2x)^2 \geq 0$$

$$(e^x - 2x)(e^x + 2x) \geq 0$$

$e^{2x} - 4x^2 \geq 0$



④

$$\left[1 + \left(\frac{dy}{dx} \right)^2 \right]^2 = 3 \frac{d^3 y}{dx^3}$$

$$1 + \left(\frac{dy}{dx} \right)^4 + 2 \left(\frac{dy}{dx} \right)^2 = 3 \frac{d^3 y}{dx^3}$$

$$3 \left(\frac{d^3 y}{dx^3} \right) - \left(\frac{dy}{dx} \right)^4 - 2 \left(\frac{dy}{dx} \right)^2 - 1 = 0$$

order = highest derivative = 3

degree = power of the highest derivative = 1 //

⑤

$$\textcircled{i} \quad x \ln x \frac{dy}{dx} + y = 25 \ln x$$

$$\frac{dy}{dx} + \frac{y}{x \ln x} = \frac{25 \ln x}{x \ln x}$$

$$\frac{dy}{dx} + \frac{y}{x \ln x} = \frac{25}{x}$$

$$\frac{dy}{dx} + \left(\frac{1}{x \ln x} \right) y = \left(\frac{25}{x} \right)$$

in the form,

$$\frac{dy}{dx} + P(x) y = Q(x)$$

This is first order linear D.E.

$$\begin{aligned}\text{Integrating factor} &= e^{\int p(x) dx} \\ &= e^{\int \frac{1}{x \ln x} dx} \\ &= e^{\int \frac{(1/x)}{\ln x} dx}\end{aligned}$$

$$\mu = e^{\ln|\ln x|}$$

$$\log_e \mu = \log_e e^{\ln|\ln x|}$$

$$\ln \mu = \ln|\ln x|$$

$$\mu = |\ln x| = \ln x //$$

(ii)

$$x \frac{dy}{dx} + y = y^2 x^2 \ln x$$

$$\frac{dy}{dx} + \frac{y}{x} = y^2 x \ln x$$

$$\frac{dy}{dx} = y^2 x \ln x - \frac{y}{x}$$

~~$$\frac{dy}{dx} = \frac{(x^2 y^2 \ln x - y)}{x}$$~~

$$\frac{dy}{dx} = (x \ln x) y^2 - \left(\frac{1}{x}\right) y$$

$$\frac{dy}{dx} + \left(\frac{1}{x}\right) y = (x \ln x) y^2$$

We have first order bernoulli D.E. (non-linear)

in the form,

$$\frac{dy}{dx} + P(x)y = Q(x)y^n \text{ where } P(x) = \frac{1}{x}, Q(x) = x \ln x, n=2$$

$$\text{Let, } Z = y^{1-2}$$

$$Z = y^{-1}$$

$$\frac{dz}{dx} = -1 y^{-2} \frac{dy}{dx}$$

$$-y^{-2} \frac{dy}{dx} = \frac{dz}{dx}$$

$$\frac{dy}{dx} + \frac{1}{x} y = (x \ln x) y^2$$

multiply the above standard form by, $-y^{-2}$,

$$\underbrace{-y^{-2} \frac{dy}{dx}} - \frac{1}{x} \times \frac{1}{y^2} y = (x \ln x) \times y^2 \times \frac{-1}{y^2}$$

$$\frac{dz}{dx} - \frac{1}{xy} = -(x \ln x)$$

$$\frac{dz}{dx} + (x \ln x) = \frac{1}{xy}$$

$$\frac{dz}{dx} + (x \ln x) = \frac{1}{x} \times Z$$

$$\frac{dz}{dx} + \left(-\frac{1}{x}\right) Z = (-x \ln x)$$

$$\text{in the form, } \frac{dz}{dx} + \underbrace{\left(-\frac{1}{x}\right)}_{P(x)} Z = \underbrace{(-x \ln x)}_{Q(x)}$$

$$\text{I.F.} = e^{\int P(x) dx}$$

$$= e^{\int -\frac{1}{x} dx}$$

$$\text{I.F.} = -\ln x //$$

$$t = \int -\frac{1}{x} dx$$

$$t = -\int \frac{1}{x} dx$$

$$t = -\ln|x|$$

(f)

02

$$a) \quad y = \frac{1}{a+x}$$

$$\frac{dy}{dx} = \frac{-1 \times (1)}{(a+x)^2}$$

$$\frac{dy}{dx} = \frac{-1}{(a+x)^2}$$

$$y = \frac{1}{a+x}$$

$$ay + xy = 1$$

$$a = \frac{(1 - xy)}{y}$$

$$a = \frac{1}{y} - x$$

$$\frac{dy}{dx} = \frac{-1}{\left(\frac{1}{y} - x + x\right)^2}$$

$$\frac{dy}{dx} = \frac{-1}{\frac{1}{y^2}}$$

$$\frac{dy}{dx} = -y^2$$

replace $\frac{dy}{dx}$ with $-1/(dy/dx)$

$$\frac{-1}{(dy/dx)} = -y^2$$

$$\frac{dy}{dx} = y^{-2}$$

$$dy = y^{-2} dx$$

$$y^2 dy = 1 dx$$

~~$$1 dx = y^2 dy = 0$$~~

$$\int y^2 dy = \int 1 dx$$

$$\frac{y^3}{3} = x + C$$

(C arbitrary constant)

$$y^3 = 3x + C //$$

(b) (i) $(D^3 + 25D)y = \sin 5x + e^{5x}$

$$(D^3 + 25D)y = 0$$

$y = e^{mx}$ is the solution.

$$Dy = m e^{mx}$$

$$D^2 y = m^2 e^{mx}$$

$$D^3 y = m^3 e^{mx}$$

$$m^3 e^{mx} + 25 e^{mx} = 0$$

$$e^{mx} (m^3 + 25m) = 0$$

$$\forall x, e^{mx} > 0$$

$$m^3 + 25m = 0$$

$$m(m^2 + 25) = 0$$

$$m = 0, m^2 - 5^2 i^2 = 0$$

$$m = 0, (m - 5i)(m + 5i) = 0$$

$$m = 0, m = 5i, m = -5i$$

complementary function

$$y = A e^{0x} + B e^{5ix} + C e^{-5ix}$$

$$y = A + B e^{5ix} + C e^{-5ix}$$

$$y = A + B(\cos 5x + i \sin 5x) + C(\cos 5x - i \sin 5x)$$

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C.F.
m

$$y = A + (B+C) \cos 5x + (B-C) i \sin 5x$$

$$y = A + B \cos 5x + C i \sin 5x$$

(A, B, C are arbitrary constants)

particular integral

$$y = \frac{1}{(D^3 + 25D)} (\sin 5x + e^{5x})$$

$$y = \underbrace{\frac{1}{(D^3 + 25D)} \sin 5x}_{\alpha} + \underbrace{\frac{1}{(D^3 + 25D)} e^{5x}}_{\beta}$$

$\alpha \rightarrow$

$$\frac{1}{(D^3 + 25D)} \sin 5x = \frac{1}{D(D^2 + 5^2)} \sin 5x$$

$$= \frac{1}{D} \frac{1}{(D^2 + 5^2)} \sin 5x$$

$$= \frac{1}{D} \text{ imaginary part of } \frac{1}{(D^2 + 5^2)} e^{i5x}$$

$$= \frac{1}{D} \text{ imaginary part of } e^{i5x} \frac{1}{(D^2 + 5^2)} e^{0x}$$

$$= \frac{1}{D} \text{ imaginary part of } e^{i5x} \frac{1}{(D - 5i)(D + 5i)} e^{0x}$$

$$= \frac{1}{D} \text{ imaginary part of } e^{i5x} \frac{1}{-5i \times 5i} e^{0x}$$

$$= \frac{1}{D} \text{Im} \left(e^{i5x} \times \frac{1}{-25i^2} \right) = \frac{1}{D} \text{Im} \left(\frac{\cos 5x + i \sin 5x}{25} \right)$$

$$\text{PARTIAL} = \frac{1}{D} \frac{\sin 5x}{25}$$

$$\int \frac{\sin 5z}{25} dz = \frac{1}{25} \frac{-\cos 5z}{5}$$

$$= -\frac{1}{125} \cos 5z \quad \text{--- } \textcircled{\alpha}$$

$\beta \Rightarrow$

$$\frac{1}{(D^3 + 25D)} e^{5z} = \frac{1}{D} \frac{1}{(D^2 + 5^2)} e^{5z}$$

$$= \frac{1}{D} \frac{1}{(D-5i)(D+5i)} e^{5z}$$

$$= \frac{1}{D} \frac{1}{(5-5i)(5+5i)} e^{5z}$$

$$= \frac{1}{D} \frac{1}{5^2 - (5i)^2} e^{5z}$$

$$= \frac{1}{D} \frac{1}{(25+25)} e^{5z}$$

$$= \frac{1}{D} \frac{1}{50} e^{5z}$$

$$= \int \frac{1}{50} e^{5z} dz$$

$$= \frac{1}{50} \int e^{5z} dz$$

$$\frac{1}{(D^3 + 25D)} e^{5z} = \frac{1}{50} \frac{e^{5z}}{5} = \frac{1}{250} e^{5z} \quad \text{--- } \textcircled{\beta}$$

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By α , β ,

particular integral

P.I.
m

$$y = -\frac{1}{125} \cos 5x + \frac{1}{250} e^{5x}$$

General solution = P.I. + C.F.

$$y = A + B \cos 5x + C i \sin 5x + \frac{-1}{125} \cos 5x + \frac{1}{250} e^{5x}$$

$$y = A + B \cos 5x + C i \sin 5x + \frac{1}{250} e^{5x} //$$

$$(ii) \quad x^2 \frac{d^2 y}{dx^2} - 3x \frac{dy}{dx} + 4y = 4 \ln x$$

we have non homogeneous but cauchy - Euler equation.

We can obtain solution in the form of, $x = e^z$

$$\ln x = z$$

$$\text{then, } \frac{dz}{dx} = \frac{1}{x}$$

$$\frac{dy}{dx} = \frac{dy}{dz} \times \frac{dz}{dx}$$

$$\frac{dy}{dx} = \frac{dy}{dz} \times \frac{1}{x}$$

$$\text{Let } \frac{d}{dx} = D, \quad \frac{d}{dz} = \mathcal{D}$$

then,

$$x \mathcal{D} y = \mathcal{D} y$$

$$x^2 D^2 y = x(x-1)y$$

$$x^3 D^3 y = x(x-1)(x-2)y$$

$$\therefore x^2 \frac{d^2 y}{dx^2} - 3x \frac{dy}{dx} + 4y = 4 \ln x$$

↓

$$x(x-1)y - 3xy + 4y = 4 \ln(e^z)$$

$$x(x-1)y - 3xy + 4y = 4 \log_e e^z$$

$$y [x(x-1) - 3x + 4] = 4z$$

$$y [x^2 - x - 3x + 4] = 4z$$

$$y [x^2 - 4x + 4] = 4z$$

$$(x^2 - 4x + 4)y = 4z$$

complementary function.

$$(x^2 - 4x + 4)y = 0$$

Let

$$y = e^{mx}$$

$$xy = m e^{mx}$$

$$x^2 y = m^2 e^{mx}$$

$$m^2 e^{mx} - 4m e^{mx} + 4 e^{mx} = 0$$

$$e^{mx} (m^2 - 4m + 4) = 0$$

$$\forall x \in \mathbb{R}, e^{mx} > 0$$

$$\therefore (m^2 - 4m + 4) = 0$$

$$(m-2)^2 = 0$$

$$m=2, m=2$$

∴ complementary function

$$y = Ae^{2x} + Be^{-2x} \quad (A, B \text{ are arbitrary constants})$$

$$y = Ax^2 + Bx^0$$

$$y = Ax^2 + (\ln x) Bx^2$$

Let's find particular integral

$$y = \frac{1}{(x^2 - 4x + 4)} \cdot 4x$$

$$y = \frac{1}{\left[4\left(\left(\frac{x^2}{4} - x\right) + 1\right)\right]} \cdot 4x$$

$$y = \frac{1}{\left(\frac{x^2}{4} - x\right) + 1} \cdot x$$

$$y = \frac{1}{1 + \left(\frac{x^2}{4} - x\right)} \cdot x$$

$$y = \left[1 + \left(\frac{x^2}{4} - x\right) + \left(\frac{x^2}{4} - x\right)^2 + \dots\right] \cdot x$$

$$y = \left[x + \frac{x^2}{4}x - x^2 + \left(\frac{x^4}{16} + x^2x - \frac{x^3}{2}\right) + \dots\right]$$

$$y = [x + 0 - 0 + 0 + 0 - 0 + \dots]$$

$$y = x - 1$$

General solution = C.F. + P.I.

$$y = Ax^2 + (\ln x) Bx^2 + (z-1)$$

$$y = Ax^2 + (\ln x) Bx^2 + (\ln x - 1)$$

(A, B are arbitrary constants) //

(C)

$$a \neq 0$$

$$\frac{dx}{dt} = -ay$$

$$\frac{dy}{dt} = ax$$

$$\text{Let } \frac{d}{dt} = \mathcal{D}$$

then,

$$\mathcal{D}x = -ay \text{ --- (1)}$$

$$\mathcal{D}y = ax \text{ --- (2)}$$

(1),

$$y = -\frac{1}{a} \mathcal{D}x \text{ --- (3)}$$

(3), differentiate w.r.t. "t".

$$\frac{dy}{dt} = -\frac{1}{a} \frac{d}{dt} \left(\frac{dx}{dt} \right)$$

$$\mathcal{D}y = -\frac{1}{a} \mathcal{D}^2x \text{ --- (4)}$$

(2) = (4),

$$ax = -\frac{1}{a} \mathcal{D}^2x$$

$$-a^2x = \mathcal{D}^2x$$

$$\mathcal{D}^2x + a^2x = 0$$

$$(\mathcal{D}^2 + a^2)x = 0$$

Let, $x = e^{mt}$ be the solution,

$$\mathcal{D}x = me^{mt}, \mathcal{D}^2x = m^2e^{mt}$$

$$\therefore m^2e^{mt} + a^2e^{mt} = 0$$

$$e^{mt}(m^2 + a^2) = 0$$

Since $e^{mt} > 0 \rightarrow e^{mt} \neq 0$

$$(m^2 + a^2) = 0$$

$$(m - ai)(m + ai) = 0$$

$$m = ai, m = -ai$$

$$y = Ae^{iat} + Be^{-iat} \quad (A, B \text{ are arbitrary constants})$$

$$y = A(\cos at + i \sin at) + B(\cos at - i \sin at)$$

$$y = (A+B) \cos(at) + (A-B)i \sin(at)$$

$$y = A \cos(at) + Bi \sin(at) //$$

②) x,

$$Dy = ax$$

$$D(A \cos at + Bi \sin at) = ax$$

$$(A \times -\sin at \times a) + (Bi \cos at \times a) = \cancel{a} \cancel{dx} \cancel{dx} ax$$

$$-A \sin at + Bi \cos at = ax$$

$$x = -A \sin(at) + Bi \cos(at) //$$

$$x^2 = A^2 \sin^2(at) + B^2 i^2 \cos^2(at) - 2ABi \sin(at) \cos(at) \text{ ——— } (\alpha)$$

$$y^2 = A^2 \cos^2(at) + B^2 i^2 \sin^2(at) + 2ABi \sin(at) \cos(at) \text{ ——— } (\beta)$$

$$\textcircled{\alpha + \beta},$$

$$x^2 + y^2 = A^2 \sin^2(at) - B^2 \cos^2(at) - 2ABi \sin(at) \cos(at) + (A^2 \cos^2(at) - B^2 \sin^2(at) + 2ABi \sin(at) \cos(at))$$

$$= A^2(\sin^2 at + \cos^2 at) - B^2(\sin^2 at + \cos^2 at)$$

$$x^2 + y^2 = \underbrace{A^2 - B^2}_{\text{constant}}$$

$$\text{Let, } A^2 - B^2 = r^2$$

then,

$$x^2 + y^2 = r^2$$

So that, the solution set represent a circle which radius is $A^2 - B^2$ and the center point of the circle is the $(0, 0)$.

03

(a) (i)

$$\frac{d^2 y}{dx^2} + 5(\ln x) \frac{dy}{dx} = 0$$

$$\frac{d^2 y}{dx^2} + 5(\ln x) \frac{dy}{dx} + 0 y = 0$$

$$P(x) = 5(\ln x), \quad Q(x) = 0$$

$$P(x) = 5 \left[\frac{1}{1}(x-1) + \frac{1}{2}(x-1)^2 + \frac{1}{3}(x-1)^3 + \frac{1}{4}(x-1)^4 + \dots \right]$$

$$P(x) = 5(x-1) \left[1 + \frac{1}{2}(x-1) + \frac{1}{3}(x-1)^2 + \frac{1}{4}(x-1)^3 + \dots \right]$$

$P(x)$ and $Q(x)$ both are analytic at $x=0$

$\therefore x=0$ is an ordinary point. //

(ii)

$$2x^2 \frac{d^2 y}{dx^2} - x \frac{dy}{dx} + (x-5)y = 0$$

$$\frac{d^2 y}{dx^2} \left(-\frac{1}{2x} \right) \frac{dy}{dx} + \left(\frac{x-5}{2x^2} \right) y = 0$$

$P(x)$
 $Q(x)$

$$P(x) = -\frac{1}{2x}, \quad Q(x) = \frac{(x-5)}{2x^2}$$

neither of these two functions are analytic at $x=0$

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$$x P(x) = \frac{-1}{2x} \times x$$

$$x P(x) = -\frac{1}{2} \quad \left. \vphantom{\frac{-1}{2}} \right\} x=0 \text{ finite}$$

$$x^2 Q(x) = \frac{(x-5)}{2} \quad \left. \vphantom{\frac{(x-5)}{2}} \right\} x=0 \text{ finite}$$

$\therefore x=0$, regular singular point. //

(b)

$$z = a e^{-by} \cos bx \text{ --- (1)}$$

$$z = z(x, y)$$

①, partially differentiate w.r.t. y

$$\frac{\partial z}{\partial y} = a \cos bx \cdot e^{-by} \cdot x - b^2$$

$$\frac{\partial z}{\partial y} = -ab^2 e^{-by} \cos bx \text{ --- (2)}$$

②, partially differentiate w.r.t. x

$$\frac{\partial z}{\partial x} = a e^{-by} \cdot x - \sin bx \cdot b$$

$$\frac{\partial z}{\partial x} = -ab e^{-by} \sin bx \text{ --- (3)}$$

③, partially differentiate w.r.t. x

$$\frac{\partial^2 z}{\partial x^2} = -ab e^{-by} \cos bx \cdot b$$

$$\frac{\partial^2 z}{\partial x^2} = -ab^2 e^{-by} \cos bx \text{ --- (4)}$$

$$\textcircled{2} = \textcircled{4},$$

$$\frac{\partial z}{\partial y} = \frac{\partial^2 z}{\partial x^2} //$$

$$(c) \quad D = \frac{\partial}{\partial x}, \quad D' = \frac{\partial}{\partial y}$$

$$(D^2 - 2DD' + D'^2)Z = 2x^3$$

~~Ans~~

~~Ans~~

C.F.
m

$$\text{Let } D = m, \quad D' = 1$$

$$(m^2 - 2m + 1) = 0$$

$$(m-1)^2 = 0$$

$$m = 1, m = 1$$

$$Z = \phi(y + mx)$$

$$Z = \phi_1(y+x) + x(y+x)\phi_2 \quad (\phi_1, \phi_2 \text{ arbitrary constants})$$

functions

P.I.
m

$$Z = \frac{1}{(D^2 - 2DD' + D'^2)} 2x^3$$

$$Z = \frac{1}{D^2 \left(1 - \left(\frac{2D'}{D} - \frac{D'^2}{D^2} \right) \right)} 2x^3$$

$$Z = \frac{1}{D^2} \left[1 + \left(\frac{2D'}{D} - \frac{D'^2}{D^2} \right) + \left(\frac{2D'}{D} - \frac{D'^2}{D^2} \right)^2 + \dots \right] 2x^3$$

$$Z = \frac{1}{D^2} \left[2x^3 + 2 \frac{D'(2x^3)}{D} - \frac{D'^2(2x^3)}{D^2} + \dots \right]$$

$$Z = \frac{1}{D^2} \left[2x^3 + 0 - 0 + \dots \right]$$

$$Z = \frac{1}{D^2} 2x^3$$

$$Z = \int 2x^3 dx = 2 \frac{x^4}{4} = \frac{x^4}{2}$$

General solution

$$Z = \phi_1(y+x) + x(y+x)\phi_2 + \frac{x^4}{2} //$$